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Microprocessors and Microcontrollers Laboratory
CSE - 2104

A project report on

"CropCare Bluetooth Rover"

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1. Introduction

Agriculture forms the backbone of Bangladesh's economy, yet farmers continue to face enormous challenges in protecting their crops from pests and insects. Conventional manual spraying of insecticides is time-consuming, physically exhausting, and exposes farmers to hazardous chemicals. The CropCare Bluetooth Rover is a low-cost, microcontroller-based agricultural rover designed to address this problem by automating the delivery of insecticide across crop fields.

Built around an Arduino Uno R3 microcontroller and an HC-05 Bluetooth module, this rover can be wirelessly controlled from a smartphone using the SriTu Hobby app. A 5V mini water pump mounted on the chassis dispenses insecticide as the rover moves across the field, covering ground efficiently without any human physical contact with chemicals. The system demonstrates core embedded system concepts including wireless serial communication, DC motor driver interfacing, real-time command execution, and pump control through a transistor switching circuit — all assembled on an affordable 2WD chassis powered by a LiPo battery and AAA pencil cells.

2. Project Idea & Objective

The core idea of this project is to build a small, wirelessly operated agricultural rover capable of navigating a crop field and spraying insecticide on command. The operator uses on-screen direction buttons in the SriTu Hobby Android app to drive the rover in four directions; pressing the Right button simultaneously activates the water pump to spray insecticide. This allows the rover to rotate in place while dispensing chemicals, covering surrounding plants in a full circle without repositioning.

Main Objectives:

- **Design and build a functional Bluetooth-controlled agricultural rover** using Arduino Uno R3.
- **Establish wireless serial communication** between a smartphone and the Arduino via the HC-05 module.
- **Implement real-time motor control** using the L298N dual H-Bridge motor driver for four-directional movement.
- **Integrate a 5V water pump** controlled through a transistor switching circuit to spray insecticide on crop fields.
- **Minimize farmer exposure to harmful chemicals** by enabling remote hands-free pesticide application.
- **Demonstrate a practical embedded system project** with real agricultural applicability at an affordable cost.

3. Project Description

The CropCare Bluetooth Rover is a wireless, Bluetooth-controlled agricultural robot designed to spray insecticide across crop fields. The rover is built on a 2WD plastic chassis featuring two rear DC gear motors for propulsion and a front directional (castor) wheel for balance and smooth turning.

The farmer installs the SriTu Hobby Android app on their smartphone and pairs it with the HC-05 Bluetooth module attached to the Arduino Uno R3. By pressing direction buttons on the app screen, single-character commands are sent over Bluetooth serial communication: 'U' for Forward, 'D' for Backward, 'L' for Left turn, 'S' for Stop, and 'R' for Right with pump activation. The Arduino receives each character, decodes the command, and drives the L298N motor driver to execute the corresponding wheel movement.

When the 'R' (Right) command is received, the Arduino not only steers the rover right but also triggers the 5V water pump through an NPN transistor and 1k ohm base resistor — activating the insecticide spray. The rover can be positioned at the center of a crop area and rotated in place while pumping, ensuring 360-degree chemical coverage without the farmer needing to physically enter the field. The system is powered by a LiPo 900mAh battery for the motors and a set of 8 AAA pencil batteries for the Arduino and logic components.

4. Key Features

- **Wireless Bluetooth Control:** Full rover control from up to 10 meters using the SriTu Hobby Android app — no physical wires required.
- **Integrated Insecticide Spraying:** A 5V water pump activates with the Right command to spray insecticide; all other commands keep the pump OFF, preventing wastage.
- **Four-Directional Movement:** Supports Forward, Backward, Turn Left, Turn Right, and Stop commands for full field maneuverability.
- **Transistor-Switched Pump Circuit:** A single NPN transistor and 1k ohm resistor form a reliable switch, protecting the Arduino's output pin from pump current draw.
- **Dual Power Supply Design:** LiPo 900mAh battery powers high-current motors; AAA pencil batteries power the Arduino and logic for a stable, noise-free control system.
- **2WD Chassis with Front Castor Wheel:** Lightweight and compact design for easy navigation on flat crop field terrain.
- **Real-Time Response:** Instant motor and pump response upon receiving each Bluetooth command with negligible latency.
- **Low-Cost Agricultural Solution:** All components are locally available in Bangladesh at a very affordable price, making this viable for small-scale farmers.

5. Components Used

Component	Specification / Role	Qty
Arduino Uno R3	ATmega328P — Main microcontroller; processes all Bluetooth commands and controls motor & pump outputs	1
HC-05 Bluetooth	2.4 GHz module — wireless serial link between SriTu Hobby app and Arduino at 9600 baud	1
L298N Motor Driver	Dual H-Bridge — controls direction of both DC motors independently	1
DC Gear Motor	Drives left and right wheels for all directional movement	2
2WD Car Chassis	Lightweight frame with front directional (castor) wheel, holds all modules	1
Drive Wheels	Rear drive wheels mounted on the DC gear motors	2
LiPo 900mAh Battery	Main rechargeable power source for motors and motor driver	1
AAA Pencil Batteries	1.5V each (used in clocks) — powers Arduino and low-current components	8
5V Water Pump	Mini submersible pump — dispenses insecticide when activated	1
Transistor (NPN)	Switches the water pump ON/OFF via Arduino digital pin signal	1
1k Ohm Resistor	Current-limiting resistor for the transistor base drive	1
Breadboard	Prototyping board for circuit connections	1
Jumper Wires	M-M and M-F wires for all component connections	Set

6. Working Principle

The CropCare Bluetooth Rover operates on wireless serial communication between the SriTu Hobby smartphone app and the Arduino Uno R3 via the HC-05 Bluetooth module. When the user presses a button on the app, a single ASCII character is transmitted over Bluetooth at 9600 baud to the HC-05 module's RX/TX serial pins, which relays it directly to the Arduino's hardware serial interface.

The Arduino continuously checks for incoming serial data using Serial.available(). Upon receiving a character, it executes the corresponding motor and pump logic. The L298N motor driver receives HIGH/LOW signals on its IN1–IN4 input pins to control the direction of both DC gear motors. The water pump is connected to pin D6 on the Arduino through a transistor circuit — when pin D6 goes HIGH, the transistor saturates and completes the pump circuit, activating the spray.

Decision Logic (SriTu Hobby App Commands):

- Command 'U' received → Both motors run forward → Rover moves forward → Pump OFF
- Command 'D' received → Both motors run in reverse → Rover moves backward → Pump OFF
- Command 'L' received → Left motor reverse, Right motor forward → Rover turns left → Pump OFF
- Command 'S' received → Both motors stop → Rover halts → Pump OFF
- Command 'R' received → Left motor forward, Right motor reverse → Rover turns right → Pump ON (insecticide spray)

The transistor (NPN type) acts as a digital switch for the pump. When Arduino pin D6 outputs 5V HIGH, current flows through the 1k ohm base resistor into the transistor base, causing collector-emitter saturation and completing the pump's power circuit from the LiPo battery. When D6 is LOW, the transistor is in cut-off mode and the pump remains off.

7. System Block Diagram

SMARTPHONE (SriTu Hobby App)	→	**HC-05 Bluetooth Module**	→	**ARDUINO UNO R3 (Main Controller)**	→	**L298N Motor Driver**	→	**Motor L & Motor R**	→	**Water Pump (5V via Transistor)**

The smartphone app sends characters to the HC-05 Bluetooth module, which forwards them to the Arduino via serial communication. The Arduino processes each character and drives the

L298N motor driver for wheel movement, while simultaneously controlling the 5V water pump via a transistor switching circuit.

8. System Flow Chart:

START

▼

Power ON → Arduino Initializes & HC-05 Begins Listening

▼

User Opens SriTu Hobby App on Smartphone

▼

Bluetooth Pairing Established (9600 baud)

▼

User Presses a Direction Button in App

▼

App Sends Single Character Command via Bluetooth

▼

Arduino Reads Incoming Serial Character

▼

Command Matching (U / D / L / R / S)?

▼

'U' → Forward() | 'D' → Backward() | 'L' → Left() | 'S' → Stop()

▼

'R' → Right() + PumpON() — Water Pump Activated

▼

L298N Motor Driver Receives PWM Signals → Drives Both Motors

▼

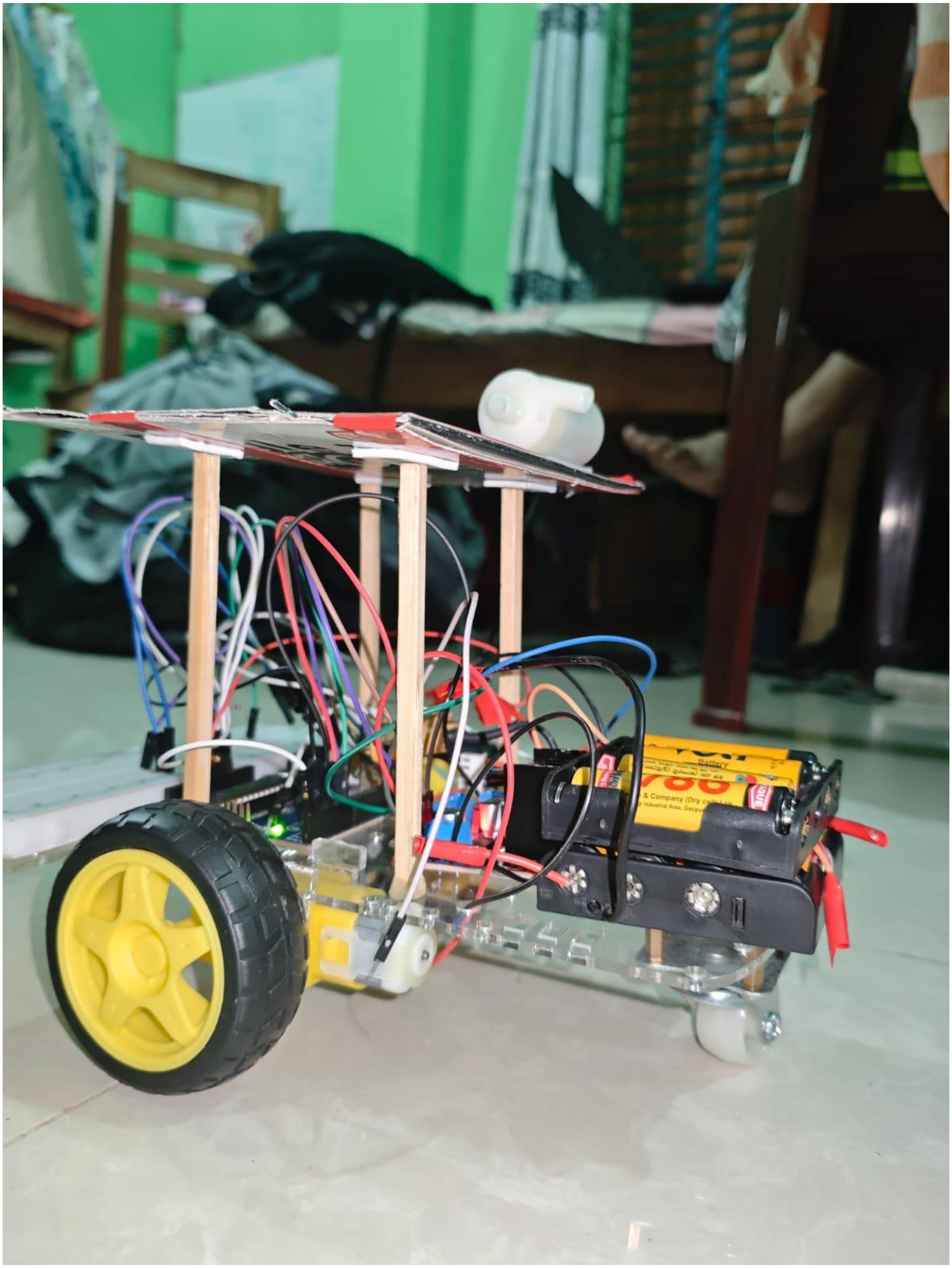
Rover Moves in Commanded Direction / Pump Dispenses Insecticide

▼

Any New Command? → Loop Back to Listening

▼

END / Power OFF


Battery

9. Applications in Real Life

The CropCare Bluetooth Rover concept addresses a wide range of practical needs in agriculture and beyond:

- **Crop Field Insecticide Spraying (Bangladesh):** The most direct application — farmers in rural Bangladesh can use this rover to safely spray insecticides on paddy, vegetable, and jute fields without physically entering the field or inhaling chemicals.
- **Greenhouse and Nursery Management:** The rover can be adapted to navigate greenhouse rows and spray pesticides or nutrients on plants in confined indoor farming environments.
- **Fertilizer Distribution:** By replacing the insecticide with liquid fertilizer, the same rover can be used for precision liquid nutrient application directly to crop roots.
- **Weed Control:** A herbicide-based version could target specific field sections for chemical weed control, reducing manual labor.
- **Disaster & Hazardous Area Access:** The basic rover design can be repurposed for exploring areas hazardous to humans, such as chemically contaminated zones or post-disaster terrain.
- **Education & Embedded Systems Training:** This project teaches students wireless communication, motor control, transistor switching, and real-time command execution — essential skills for careers in robotics, IoT, and embedded systems.
- **Small Factory Floor Monitoring:** With minor modifications, the chassis can carry sensors to monitor environmental conditions in small industrial units or garment factories across Bangladesh.

10. Limitations

- **Limited Bluetooth Range:** The rover can only be controlled within approximately 10 meters of Bluetooth range, which restricts usability across large open crop fields or in areas with signal obstructions.
- **No Terrain Adaptability:** The 2WD plastic chassis and small wheels are designed for flat surfaces. Muddy, uneven, or wet agricultural terrain common in Bangladesh can impede the rover's movement.
- **Manual Control Only:** The rover has no autonomous navigation capability. The operator must manually guide it throughout the entire spraying session, requiring constant attention.
- **Limited Battery Life:** The LiPo 900mAh battery and AAA cells provide a restricted operating duration. There is no battery level indicator, risking sudden shutdown during operation.
- **No Water Level Monitoring:** The system cannot detect when the insecticide reservoir is empty, requiring the operator to manually check the pump fluid level before and during use.

- **Prototype-Level Build:** As a laboratory project, the chassis and circuit assembly may lack the structural durability and weatherproofing required for prolonged outdoor agricultural use.

11. Existing / Similar Models:

- **Manual Backpack Sprayers:** Traditional knapsack sprayers used widely across Bangladesh — require the farmer to walk the field manually and are physically demanding with direct chemical exposure.
- **Tractor-Mounted Boom Sprayers:** Large-scale mechanical sprayers mounted on tractors for wide-area coverage — far more expensive and impractical for small Bangladesh farm plots.
- **Drone-Based Crop Spraying (DJI Agras):** GPS-guided agricultural drones for aerial spraying — highly effective but extremely expensive and require specialized operator licensing.
- **Arduino Bluetooth RC Car Projects:** Common hobbyist projects using Arduino Uno and HC-05 to build remote-controlled cars — the mechanical foundation this project is adapted from, without agricultural spraying capability.
- **Wi-Fi Rover with ESP8266/ESP32:** A more advanced version using Wi-Fi for greater range control through a browser or dedicated app — but at higher component cost and complexity.

12. Conclusion

The CropCare Bluetooth Rover is a practical, low-cost, and impactful embedded systems project that directly addresses a real-world agricultural challenge in Bangladesh. By combining wireless Bluetooth control, DC motor driving, and a transistor-switched water pump circuit on an Arduino Uno platform, this rover provides a working prototype for remotely operated crop spraying — significantly reducing farmer exposure to hazardous insecticides and lowering the physical burden of manual field treatment.

The project successfully demonstrates core microcontroller concepts including serial communication, H-bridge motor control, real-time command decoding, and transistor-based load switching — all within a compact and affordable hardware setup. The use of the SriTu Hobby app as the control interface provides an intuitive and immediately accessible user experience without requiring any custom app development.

While the current prototype is limited in range and terrain adaptability, its foundational design is highly upgradable. Future versions could incorporate GPS-based autonomous navigation, obstacle avoidance using ultrasonic sensors, a longer-range Wi-Fi module, and a sealed weatherproof chassis — transforming this laboratory project into a viable commercial agricultural product for Bangladesh's farming community.

13. References

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